

Control Chart:

What is a Control Chart?

A **Control Chart** is a graphical tool used to monitor the stability and variability of a process over time. It is a key component of **Statistical Process Control (SPC)**.

Purpose: Detecting any unusual variations in the process (common vs. special causes).

Components of a Control Chart

- **Center Line (CL):** The average of the data.
- **Upper Control Limit (UCL):** Usually set at +3 standard deviations from the mean.
- **Lower Control Limit (LCL):** Usually set at -3 standard deviations from the mean.
- **Data Points:** Plotted values of the measured variable or attribute over time.

Types of Control Charts

For Variables (measurable data)

- **$\bar{X}$ -Chart (Mean Chart):** Monitors the average of the process.
- **R-Chart (Range Chart):** Monitors the dispersion within subgroups.
- **S-Chart (Standard Deviation Chart):** Alternative to the R-Chart for large sample

For Attributes (count data)

- **p-Chart:** Proportion of defectives in a sample.
- **np-Chart:** Number of defectives in a fixed sample size.
- **c-Chart:** Number of defects per unit.
- **u-Chart:** Defects per unit where sample size varies.

Engaging Example: Cookie Factory Quality Control

Scenario:

A cookie factory produces chocolate chip cookies. The factory wants to ensure that each pack contains a consistent number of chocolate chips.

- Every hour, a random sample of **5 cookie packs** is selected.
- The number of chocolate chips in each pack is counted.

Hour	Pack 1	Pack 2	Pack 3	Pack 4	Pack 5	$\bar{X}$ (Mean)	R = max-min (Range)
1	25	27	24	28	26	26	4
2	26	25	24	25	26	25.2	2
3	27	30	29	28	29	28.6	3
4	22	23	24	22	23	22.8	2
5	26	26	26	27	25	26	2
6	25	25	27	29	30	27.2	5
					mean	25.97	3

Step-by-Step: $\bar{X}$ and R Control Chart

1. Calculate Overall Average $\bar{\bar{X}}$ and $\bar{R}$:

$$\bar{\bar{x}} = (26+25.2+28.6+22.8+26.0+27.2)/6 = 25.97$$

$$\bar{R} = (4+2+3+2+2+5)/6=3$$

Use **A2 constant**(for sample size(n)=5, **A2 = 0.577**) to calculate

$$1) \quad \bar{\bar{x}} = (26+25.2+28.6+22.8+26.0+27.2)/6 = 25.97 \text{ (middle line)}$$

$$2) \quad UCL (\bar{x}) = \bar{\bar{x}} + A2 * \bar{R} = 25.96 + 0.577 * 3 = 27.7 \text{ (upper line)}$$

$$3) \quad LCL (\bar{x}) = \bar{\bar{x}} - A2 * \bar{R} = 25.97 - 0.577 * 3 \approx 24.2 \text{ (Lower line)}$$

Here:

- $\bar{\bar{x}}$ = average of the sample means
- $\bar{R}$ = average of the sample ranges
- **A2** adjusts for the sample size.

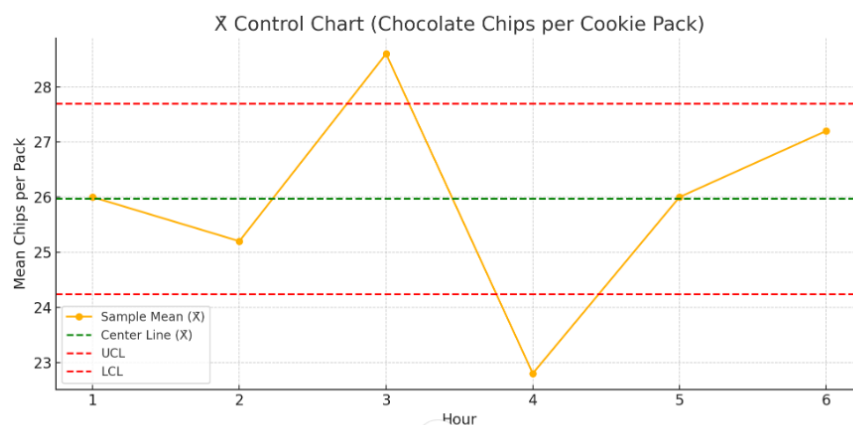
A2 is a constant used to compute the **Upper and Lower Control Limits (UCL and LCL)** for an $\bar{X}$ (mean) chart, when the **range (R)** is used as a measure of variability.

Why Do We Use Constants Like A2?

They are derived from statistical theory assuming a normal distribution and are based on the relationship between standard deviation and range. These constants allow us to set control limits that reflect **±3 sigma** control (99.73% confidence) without calculating standard deviation directly.

Plot the sample means and ranges on control charts.

- Check if all points fall within UCL and LCL.
- Look for **patterns** (e.g., seven points in a row trending upward = special cause).
- If all points are within control limits and show no unusual pattern → **Process is in control**.



Here's the $\bar{X}$ (mean) control chart based on the cookie factory example. You can observe:

- The **green dashed line** is the overall average (center line).

- The **red dashed lines** are the upper and lower control limits (UCL and LCL).
- All data points fall within the control limits, so the process appears to be **in control**

Table for A2 for different sample size

Sample Size (n)	A2 Value
2	1.88
3	1.023
4	0.729
5	0.577
6	0.483
7	0.419
8	0.373
9	0.337
10	0.308

R -Chart

Formulas for R-Chart Control Limits

$$UCL_R = D4 \times \bar{R}$$

$$LCL_R = D3 \times \bar{R}$$

Where

$\bar{R}$ = average range of the sample sub group

In $\bar{R}$ – chart (range chart), D3 and D4 are constants used to calculate the

Upper Control Limit (UCL) and

Lower Control Limit (LCL)

for the range of a sample, based on sample size.

D3 and D4 are constants that depend on sample size

They come from statistical tables based on the assumption that the data follows a normal distribution.

You do not calculate D3 and D4 manually in practice — instead, you look them up based on sample size.

Table of D3 and D4 constant

Sample Size (n)	D3	D4
2	0	3.267
3	0	2.574
4	0	2.282
5	0	2.114
6	0	2.004
7	0.076	1.924
8	0.136	1.864
9	0.184	1.816
10	0.223	1.777

Plot the R-char for above data:

Sample size n=5

$\bar{R} = 3.0$ (average range from the data)

From control chart constants table:

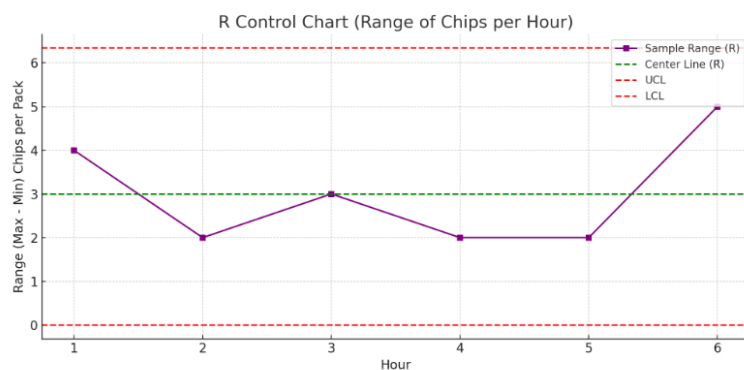
- $D3=0.000$
- $D4=2.114$

Calculation of UCL and LCL

$\bar{R} = 3.0$ (average range from the data) middle line

$UCL_R = D4 \times \bar{R} = 2.114 \times 3.0 = 6.342$ – Upper line

$LCL_R = D3 \times \bar{R} = 0.000 \times 3.0 = 0.000$ Lower line

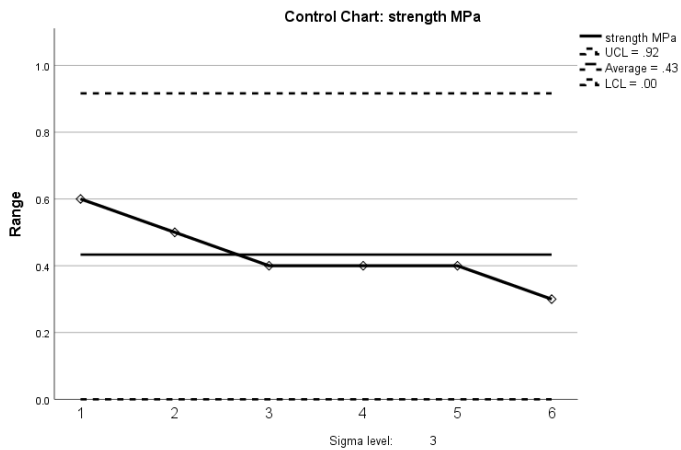
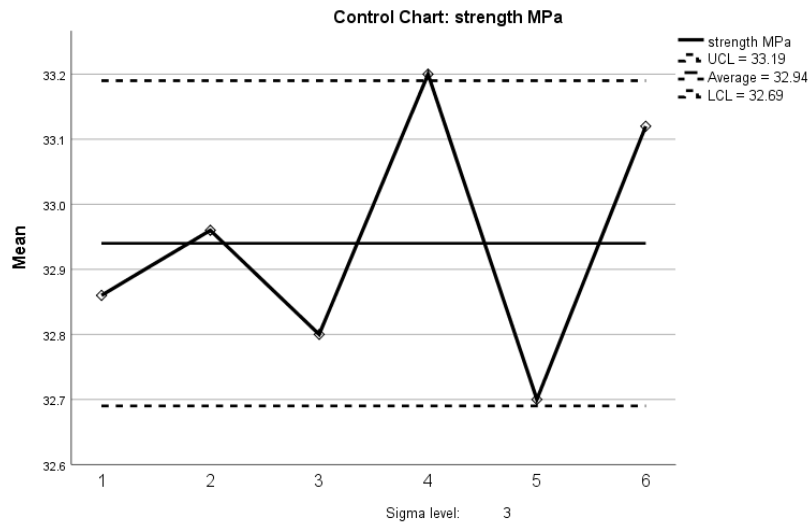


Here's the **R (Range) Control Chart** for the cookie chip counts:

- The **green dashed line** is the average range ($\bar{R}$).
- The **red dashed lines** show the upper and lower control limits.
- All points lie within limits, indicating **process variability is in control**.

A **civil engineering case**, specifically for monitoring **concrete compressive strength** across different batches using an **$\bar{X}$ and R control chart**. This is commonly used to ensure quality and consistency in construction materials.

Batch	Strength (MPa)
1	32.5
1	33.1
1	32.8
1	33.0
1	32.9
2	33.2
2	33.0
2	32.7
2	33.1
2	32.8
3	32.6
3	32.9
3	33.0
3	32.7
3	32.8
4	33.3
4	33.1
4	33.0
4	33.2
4	33.4
5	32.7
5	32.8
5	32.9
5	32.6
5	32.5
6	33.0
6	33.2
6	33.1
6	33.3
6	33.0



Example for Aeronautical Engineering:

an **aeronautical engineering** case, specifically for monitoring **engine thrust output** across different test cycles using an **$\bar{X}$ and R control chart**. This kind of chart helps ensure consistency and detect anomalies in aircraft engine performance.

Test Cycle	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
1	120.5	121.0	120.8	120.7	121.2
2	121.1	120.9	121.3	121.0	120.8

Test Cycle	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
3	120.6	120.7	120.9	121.0	120.8
4	121.2	121.3	121.1	121.0	121.4
5	120.7	120.6	120.8	120.9	120.5
6	121.0	121.2	121.1	121.3	121.0

P-chart

A p-chart (proportion chart) is a type of attribute control chart used to monitor the proportion of nonconforming (defective) items in a sample over time.

- It's appropriate when each item is classified as either defective or not defective.
- Sample sizes may vary.

Formula for the P-chart

1) Central Line(CL)

$$\bar{p} = \frac{\sum p_i}{k}$$

$\bar{p}$ = average proportion defective

n = sample size

k = number of sample

2) Standard Error $\bar{p}$

$$\sigma_p = \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

If n varies, calculate control limits for each sample individually using that sample's size n_i .

3) Control Limits

If sample size n is constant

$$UCL_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$LCL_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

If LCL is negative, set it to 0, since a proportion can't be negative.

Suppose we have 5 samples of size 100, with the following number of defective items:

Sample	Defectives (d)	Sample Size (n)	p
1	5	100	0.05
2	3	100	0.03
3	7	100	0.07
4	4	100	0.04
5	6	100	0.06

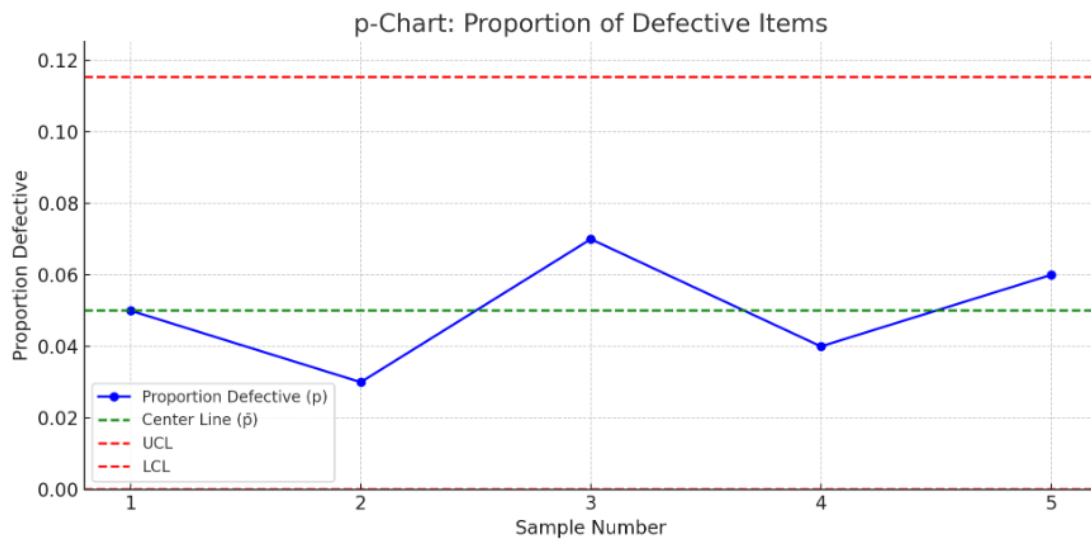
$\bar{p}$ (average pf p) = 0.05

$$\sigma_p = \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$= 0.0218$$

$$UCL = 0.05 + 3(0.0218) = 0.1154$$

$$LCL = 0.05 - 3(0.0218) = -0.0154 \Rightarrow \text{Set to 0}$$



Here's the p-chart showing the proportion of defective items in each sample:

- The blue line shows the observed defect proportions.
- The green dashed line is the average defect rate $\bar{p}$.
- The red dashed lines are the Upper and Lower Control Limits (UCL & LCL).

All points fall within control limits, so the process appears to be **in control**.